



Skin manifestations of systemic diseases:

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Skin manifestations of systemic diseases:

Learning outcomes:

- What are the common skin manifestations of systemic diseases?
- Which systemic disease has skin manifestations ?
- How we connect the skin symptoms to a systemic cause?



Skin manifestations of systemic diseases:

These are the main groups:

- Cutaneous Rheumatology.
- Cutaneous Hematology and Oncology.
- Endocrine and metabolic disorders.
- Cutaneous Gastroenterology.
- Other systemic diseases.

Skin manifestations of systemic diseases:

Cutaneous Rheumatology:

- Rheumatoid nodules: seen in ~20% of RA patients, often associated with high-titer rheumatoid factor (RF), these are firm, semi-mobile papulonodules favoring periarticular locations (e.g. elbows) and other sites of repetitive trauma or pressure.

lesions

- Rheumatoid neutrophilic dermatitis *→ histologically there's neutrophils, pain, ↑ Temp, edema, erythema*

- Sweet syndrome and pyoderma gangrenosum *→ in Rheumatoid vasculitis*

skin manifestations of systemic diseases:

- Bywaters lesions.
- Rheumatoid vasculitis: Associated with long-standing joint disease. *skin presentation may be*
- Complications of therapy for RA: NSAIDs, methotrexate, leflunomide, tocilizumab these may cause accelerated rheumatoid nodulosis – sudden appearance of multiple papulonodules, especially on the hands. Also TNF inhibitors may have local injection site reactions, urticaria, urticarial eruptions; eczematous and psoriasiform eruptions, palmoplantar pustulosis.

skin manifestations of systemic diseases:



Sweet syndrome

Skin manifestations of systemic diseases:



Rheumatoid nodules

skin manifestations of systemic diseases:



Bywaters lesions

skin manifestations of systemic diseases:

CUTANEOUS HEMATOLOGY AND ONCOLOGY:

- Disorders of mutual interest to the dermatologist and hematologist/oncologist are:
- metastatic diseases, leukemia/lymphoma, dysproteinemias, histiocytoses, vascular neoplasms, cutaneous T cell lymphoma, melanoma, and cutaneous reactions to chemotherapeutic agents. Lastly skin signs of internal malignancy.

skin manifestations of systemic diseases:

The most common skin paraneoplastic dermatosis are:

- Acquired hypertrichosis lanuginosa. *→ lanugo hair suddenly appear → pathognomonic of cancer*
- Bazex syndrome (acrokeratosis paraneoplastica) (mostly in upper aerodigestive tract malignancy). *→ eruptions → look like to chronic eczema, unresponsive to therapy*
- Carcinoid syndrome
- Ectopic adrenocorticotrophic hormone (ACTH) syndrome: (often a small cell carcinoma of the lung) *→ ACTH produce skin and hyperpigmentation*
- Paraneoplastic pemphigus *→ vesicle*

skin manifestations of systemic diseases:

- Acanthosis nigricans: (adenocarcinomas)
- Dermatomyositis (adult) *→ should sign of Gotthard Refractory periorbital telangiectasis*
- Acquired angioedema due to C1 esterase inhibitor dysfunction. *→ without urticaria is usually due to malignancy*
- Acquired ichthyosis *→ onset in Adulthood*
- Exfoliative erythroderma: (T cell lymphoma)
- Sign of Leser–Trélat: Rapid appearance or growth of multiple seborrheic keratoses. *→ all over trunk
↳ brown overgrowth on skin → stuck-on appearance*

skin manifestations of systemic diseases:



Acanthosis nigricans

skin manifestations of systemic diseases:



Sign of Leser–Trélat:

skin manifestations of systemic diseases:



Acquired hypertrichosis lanuginosa



skin manifestations of systemic diseases:

CUTANEOUS DISEASES IN PATIENTS WITH ENDOCRINOLOGIC AND/OR METABOLIC DISORDERS:

Diabetes mellitus is a very common medical condition that affects almost every organ system. It may present with these:

- Acanthosis nigricans.
- Acral dry gangrene.
- Acral erythema.
- Diabetic bullae (bullous diabeticorum)

most common in diabetic foot



skin manifestations of systemic diseases:

- Diabetic dermopathy. *subtle common*
- Disseminated granuloma annulare. *subtle annular lesion, usually in chron: DM*
- Eruptive xanthomas.
- Hemochromatosis.
- Necrobiosis lipoidica.
- Neuropathic ulcers. *tip toes, sole of foot
site of trauma*
- Scleredema.

skin manifestations of systemic diseases:



Diabetic Dermopathy

skin manifestations of systemic diseases:



Bullous diabeticorum

skin manifestations of systemic diseases:



Neuropathic ulcer



skin manifestations of systemic diseases:

metabolic syndrome are central obesity, hypertension, dyslipidemia, and type 2 diabetes.

Women with polycystic ovary syndrome have a greater frequency of metabolic syndrome and often present with acanthosis nigricans, acne vulgaris, and hirsutism.

An association between psoriasis vulgaris and metabolic syndrome has also been identified.

Thyroid disease is also associated with a significant number of dermatological disorders.



skin manifestations of systemic diseases:

- Hyperthyroidism:

The skin is fine, velvety, smooth, warm and moist due to increased sweating, also pretibial myxedema, urticaria, dermatographism, increased incidence of vitiligo.

The hair may be thin with mild, diffuse alopecia and increased incidence of alopecia areata. While the nails may present with onycholysis or koilonychia.



skin manifestations of systemic diseases:

○Hypothyroidism:

Dry, rough, coarse skin, cold and pale, boggy and edematous skin (myxedema), yellow discoloration as a result of carotenoderma.

Acquired ichthyosis and palmoplantar keratoderma, with increased incidence of vitiligo.

The hair is usually dull, coarse, brittle with slow growth, Alopecia of the lateral third of the eyebrows increased incidence of alopecia areata. The nails are thin, brittle, striated with slow growth, and onycholysis (rare).



skin manifestations of systemic diseases:

Adrenal disease classically manifests as excessive (Cushing syndrome) or insufficient (Addison disease) corticosteroid activity, this can include gluco- and mineralocorticoids. Excessive production of adrenal androgens may also occur.

Handwritten notes:
Cushing → Cushing's
 → moon-face
 → hirsutism



skin manifestations of systemic diseases:

CUTANEOUS GASTROENTEROLOGY:

Ulcerative colitis and Crohn disease:

- Erythema nodosum, urticaria, cutaneous small vessel vasculitis, pyoderma gangrenosum, peristomal lesions, Sweet Syndrome.
- oral lesions like granulomatous infiltrates (with Crohn disease only), aphthosis and angular cheilitis.
- Fistulae (perianal and abdominal), epidermolysis bullosa acquisita, psoriasis, hidradenitis suppurativa.



skin manifestations of systemic diseases:

CUTANEOUS GASTROENTEROLOGY:

Association between skin diseases and GIT bleeding:

Hereditary Hemorrhagic telangiectasia (HHT), vasculitis, ulcerative colitis, crohn disease, and kaposi sarcoma (KS).



Skin manifestations of systemic diseases:

CUTANEOUS GASTROENTEROLOGY:

Dermatologic aspects of liver disease:

- Cirrhosis: is presented with spider angiomas and other telangiectasias, palmar erythema, dilated abdominal wall veins, pruritus, and jaundice.

Hepatitis B and C viruses: Cutaneous small vessel vasculitis, cryoglobulinemic vasculitis, urticarial vasculitis, polyarteritis nodosa, necrotizing livedo reticularis, urticaria, pruritus, Lichen planus, erythema multiforme, and erythema nodosum.

Skin manifestations of systemic diseases:

OTHER SYSTEMIC DISEASES WITH CUTANEOUS MANIFESTATIONS:

- Pruritus is a feature of uremia, but can also occur in non-uremic patients on hemodialysis. *in end stage renal failure*
- Solid organ transplant recipients receive immunosuppressive agents, Will predisposes the patient to two significant cutaneous problems: infection (e.g. HPV, CMV) and malignancy (e.g. squamous cell carcinoma, Kaposi sarcoma, Merkel cell carcinoma).



Skin manifestations of systemic diseases:

- Pregnancy is an important physiologic state that is associated with its own specific dermatoses.
- Pulmonary diseases also may have skin manifestations, for example: cyanosis, nail clubbing, cutaneous metastasis of lung cancer, acanthosis nigricans, erythema gyratum repens, tripe palms, acquired ichthyosis, or acquired hypertrichosis lanuginosa.

Skin manifestations of systemic diseases:



thickened
waxy-appearing
rough surface
deepening of
skin creases

Tripe palms

Skin manifestations of systemic diseases:



Erythema gyratum repens → multicentric annular patch central scale



Skin manifestations of systemic diseases:

In conclusion; any chronic, systemic disease may affect the skin, hair and nails, making the diagnosis of the underlying disease in most of these cases a necessary step for the treatment of the skin symptoms, and furthermore, to diagnose the life threatening conditions like malignancies that may have the skin manifestations as the presenting feature.



Reference:

Bologna text book 5th edition 2024

Questions?

